

The Most-Informative-Boolean-Function Conjecture Holds for $n = 4$

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Abstract

Let X be uniform on $\{0, 1\}^n$ and let Y be the output of a memoryless binary symmetric channel with crossover probability α on input X . Courtade and Kumar conjectured that every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ satisfies $I(f(X); Y) \leq 1 - h(\alpha)$, where h is the binary entropy function. We prove the conjecture for $n = 4$, for the full range $\alpha \in (0, \frac{1}{2})$, with dictator functions the unique maximizers on $[0.001, 0.495]$. The proof is computer-assisted but self-contained: the noise range is split into four overlapping regimes; the middle regimes are certified by outward-rounded interval arithmetic over the 222 NPN equivalence classes of 4-variable Boolean functions, and the two endpoint regimes are handled by short analytic lemmas — a boundary-degree-profile bound at low noise and a χ^2 -type expansion at high noise — whose finitely many per-class hypotheses are verified in exact rational arithmetic. No prior high-noise theorem is invoked. All code (two scripts totalling ~ 350 lines, standard libraries plus `mpmath`) is available in the repository.

1 Introduction

Let $X \sim \text{Uniform}(\{0, 1\}^n)$ and let Y be obtained from X by passing each coordinate independently through $\text{BSC}(\alpha)$, $\alpha \in (0, \frac{1}{2})$. Courtade and Kumar [1] conjectured

$$I(f(X); Y) \leq 1 - h(\alpha) \quad \text{for every Boolean } f : \{0, 1\}^n \rightarrow \{0, 1\}, \quad (1)$$

with the dictator functions $f(x) = x_i$ (and their complements) achieving equality. The conjecture remains open in general; it is known in the high-noise regime $\alpha \in [\frac{1}{2} - \delta, \frac{1}{2})$ for an inexplicit absolute constant δ [2, 3], for a large explicit correlation range in the balanced case via local-optimality methods [4], and is equivalent to a symmetrized Li–Médard conjecture [5]. Two claimed proofs were withdrawn [6, 7]. Numerical verification at small n has been reported informally in the literature, but we are not aware of a proof covering a *continuum* of noise levels at any fixed $n \geq 2$; in particular, prior high-noise theorems cannot be combined with grid computations to cover all of $(0, \frac{1}{2})$, because their constants are not explicit.

Theorem 1. *For $n = 4$, inequality (1) holds for every $\alpha \in (0, \frac{1}{2})$ and every Boolean function f . Moreover, for $\alpha \in [0.001, 0.495]$, equality holds if and only if f is a dictator or an anti-dictator.*

*This result was produced by AI agents (Claude Fable 5, Anthropic; with independent verification by a second AI system) operating on the open-problems repository <https://github.com/gokhanmergen/information-theory-problems>, and reviewed by the human maintainer. Code reproducing every step is in the repository.

The interest of the result is partly the statement and partly the method: the endpoint lemmas isolate exactly the finite data that controls (1) near $\alpha \rightarrow 0$ (edge-boundary profiles; edge-isoperimetry on Q_4) and near $\alpha \rightarrow \frac{1}{2}$ (level-one Fourier weight and spectral ℓ_1 -mass), and the same four-regime scheme applies in principle at any fixed n .

2 Setup and reduction

Write $g_f(\alpha) = 1 - h(\alpha) - I(f(X); Y)$, $q_1 = \Pr[f(X) = 1] = k/16$, $q_0 = 1 - q_1$. Since Y is uniform and q_v does not depend on α ,

$$g_f(\alpha) = H(f(X), Y)(\alpha) - h(\alpha) - 3 - H(q_1),$$

and the 2×16 joint probabilities are the polynomials $p_{v,y}(\alpha) = 2^{-4} \sum_{k=0}^4 c_{v,y,k} \alpha^k (1 - \alpha)^{4-k}$ with integer counts $c_{v,y,k} = \#\{x : f(x) = v, d(x, y) = k\}$.

$I(f(X); Y)$ is invariant under permutations of the coordinates, flips of input coordinates, and complementation of the output, so it suffices to prove (1) for one representative of each NPN equivalence class. For $n = 4$ there are 222 classes (a classical count, reproduced by our enumeration, whose orbit sizes sum to 2^{16}). Constant functions give $I = 0$; dictators give $I = 1 - h(\alpha)$ exactly. The remaining 220 classes are treated by:

$\alpha \in (0, 10^{-3}]$	Lemma 1 (exact rational checks)
$\alpha \in [10^{-3}, 0.0055] \cup [0.005, 0.495]$	interval arithmetic (Section 5)
$\alpha \in [0.495, \frac{1}{2})$	Lemma 2 (exact rational checks)

3 The low-noise lemma

For $y \in \{0, 1\}^4$ let b_y be the number of Hamming neighbors x of y with $f(x) \neq f(y)$, and set $c_1 = \frac{1}{16} \sum_y b_y$ (so $16c_1 = 2B$ with B the edge boundary of $f^{-1}(1)$) and $c_2 = \frac{1}{16} \sum_y b_y \log_2 b_y$.

Lemma 1. *Let $\alpha \in (0, 10^{-3}]$, $t = \log_2(1/\alpha)$, $t_0 = \log_2 1000$, $\gamma = (1 - 10^{-3})^3$. If f is balanced and $t_0(\gamma c_1 - 1) \geq \log_2 e + c_2$, or f is unbalanced and $10^{-3}[t_0 \max(0, 1 - \gamma c_1) + \log_2 e + c_2] \leq 1 - H(q_1)$, then $g_f(\alpha) \geq 0$.*

Proof. Write $g_f = m_0 - D(\alpha)$ with $m_0 = 1 - H(q_1)$ and $D(\alpha) = h(\alpha) - H(f(X) | Y)$. Three elementary steps:

(i) $H(f(X) | Y=y) = h(\Pr[f(X) \neq f(y) | y])$, and $\Pr[f(X) \neq f(y) | y] \geq b_y \alpha (1 - \alpha)^3 =: b_y \beta$ by keeping only the distance-one terms; both quantities lie in $[0, 4\alpha] \subseteq [0, \frac{1}{2}]$ and h is increasing there, so $H(f(X) | Y) \geq \frac{1}{16} \sum_y h(b_y \beta)$.

(ii) $h(u) \geq u \log_2(1/u)$, hence $h(b_y \beta) \geq b_y \beta [t - \log_2 b_y]$ using $\beta \leq \alpha$.

(iii) $h(\alpha) \leq \alpha(t + \log_2 e)$, from $(1 - \alpha) \ln \frac{1}{1 - \alpha} \leq \alpha$.

Since $\beta \geq \gamma \alpha$ on the range, $D(\alpha) \leq \alpha[t(1 - \gamma c_1) + \log_2 e + c_2]$. If f is balanced then $m_0 = 0$ and $c_1 \geq \frac{3}{2}$: the minimum edge boundary of a balanced non-dictator subset of Q_4 is 12 (exhaustively checked over the balanced NPN classes; by Harper's theorem [8] the subcubes, i.e. dictators, are the unique sets at the minimum 8). The t -coefficient is then negative, and $t \geq t_0$ gives the stated criterion. If f is unbalanced, αt is increasing on $(0, 1/e)$, so D is maximized at $\alpha = 10^{-3}$, giving the stated criterion against $m_0 > 0$. $\square$

All 220 classes satisfy the applicable criterion; the verification uses exact rational two-sided bounds for the constants $\log_2 3, \log_2 5, \log_2 7, \log_2 11, \log_2 13, \log_2 e, \log_2 1000$, each used in the unfavorable direction.

4 The high-noise lemma

Let $\rho = 1 - 2\alpha$, $\widehat{\mathbf{1}}_f(S)$ the Fourier coefficients of the indicator, $\widetilde{W}_1 = \sum_{|S|=1} \widehat{\mathbf{1}}_f(S)^2 / (q_0 q_1)$, $A = \sum_{S \neq \emptyset} |\widehat{\mathbf{1}}_f(S)|$, $q_{\min} = \min(q_0, q_1)$.

Lemma 2. *Let $\rho_0 = 10^{-2}$. If $\rho_0 A / q_{\min} \leq \frac{1}{2}$ and*

$$[\widetilde{W}_1 + \rho_0^2(1 - \widetilde{W}_1)](1 + \frac{4}{3}\rho_0 A / q_{\min}) \leq 1,$$

then $g_f(\alpha) \geq 0$ for all $\rho \in (0, \rho_0]$.

Proof. $\Pr[f=1 \mid Y=y] = T_\rho \mathbf{1}_f(y)$, so with $p(v, y) = q_v 2^{-4}(1 + \varepsilon_{v,y})$ we get $|\varepsilon_{v,y}| \leq \rho A / q_{\min} =: \varepsilon_{\max}$ and, by Parseval and $\sum_{S \neq \emptyset} \widehat{\mathbf{1}}_f(S)^2 = q_0 q_1$,

$$\chi^2 := \sum_{v,y} q_v 2^{-4} \varepsilon_{v,y}^2 = \frac{\sum_{S \neq \emptyset} \widehat{\mathbf{1}}_f(S)^2 \rho^{2|S|}}{q_0 q_1} \leq \rho^2 [\widetilde{W}_1 + \rho^2(1 - \widetilde{W}_1)].$$

For $|\varepsilon| \leq \frac{1}{2}$, $(1 + \varepsilon) \ln(1 + \varepsilon) \leq \varepsilon + \frac{\varepsilon^2}{2} + \frac{2}{3}|\varepsilon|^3$: the series $(1 + \varepsilon) \ln(1 + \varepsilon) = \varepsilon + \sum_{k \geq 2} \frac{(-1)^k \varepsilon^k}{k(k-1)}$ is, for $\varepsilon \geq 0$, alternating with decreasing terms after $\varepsilon^2/2$ (negative leading tail term), and for $\varepsilon \in [-\frac{1}{2}, 0)$ all terms are positive with tail at most $\frac{|\varepsilon|^3}{6} \cdot \frac{1}{1-|\varepsilon|} \leq \frac{|\varepsilon|^3}{3}$. Since $\sum_{v,y} q_v 2^{-4} \varepsilon_{v,y} = 0$, summing gives $I(f(X); Y) \ln 2 \leq \frac{\chi^2}{2}(1 + \frac{4}{3}\varepsilon_{\max})$. Finally $1 - h(\alpha) = \frac{1}{\ln 2} \sum_{k \geq 1} \frac{\rho^{2k}}{2k(2k-1)} \geq \frac{\rho^2}{2 \ln 2}$ with all series terms positive, and the hypothesis is monotone in ρ on $(0, \rho_0]$. $\square$

All 220 classes pass, *including all balanced ones*, with exact dyadic Fourier data; the maximum of $\widetilde{W}_1$ over non-dictator classes is 52/63, attained by a dictator altered on a single input. (Thus no external high-noise theorem is needed; compare [2, 4].)

5 Certified interval arithmetic on the middle range

On $[10^{-3}, 0.0055] \cup [0.005, 0.495]$ we certify $g_f > 0$ for the 221 non-dictator classes by adaptive bisection in α : for each subinterval, g_f is enclosed using outward-rounded interval arithmetic (`mpmath.iv`, 90-bit precision) applied to the exact integer-coefficient polynomial joint distribution; a subinterval is discharged only when the enclosure's *infimum* is positive. Both runs complete with zero failures (2029s and 2.6s respectively, single-threaded). The trusted computing base beyond `mpmath` is a ~ 30 -line enclosure routine, printed in the repository, plus the NPN enumeration (whose class count and orbit-size sum are checked against known values).

6 Discussion

The theorem and its proof artifacts are reproducible from the repository: `endpoint_lemmas_n4.py` (exact rational checks, standard library only, <1 minute) and `certify_n4.py` (interval certification). An independent AI system re-derived both endpoint regimes by different techniques (a Taylor-remainder argument exploiting the sign of $h^{(4)}$ at high noise; certification to 10^{-6} plus Fannes–Audenaert continuity at low noise), giving two independent proofs of the endpoints.

The four-regime scheme is dimension-generic. For $n = 5$ (616,126 NPN classes) the endpoint lemmas carry over verbatim with recomputed margins; the middle-range certification is

a parallelizable but $\sim 3000\times$ larger computation. The genuinely new difficulty as n grows is only quantitative: the dictator’s isolation gap shrinks (single flips carry input mass 2^{-n}), so the certified middle range must push closer to the endpoints, where the lemmas’ margins — edge-isoperimetric at low noise, spectral at high noise — fortunately *grow* more robust.

References

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